

Springboard Virtual Internship 6.0

PROJECT INSIGHTS REPORT

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Branch: ELECTRICAL AND ELECTRONICS ENGINEERING

Domain: DATA VISUALIZATION

Project: VISUALIZING U.S. NATURAL DISASTER DECLARATIONS

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Visualizing US Natural Disaster Declarations: Trends and Patterns

1. Overview of Insights

- This report summarizes the key insights derived from the analysis of U.S. natural disaster declarations.
- The insights are based on temporal, geographic, and incident type analysis performed across all milestones.
- The objective of this report is to identify patterns, trends, and relationships in disaster occurrences and provide a clear understanding of disaster behavior over time and across regions.

2. Problem Statement.

The increasing frequency and impact of natural disasters in the United States require proper analysis to understand patterns and trends. However, raw disaster data from FEMA is complex and difficult to interpret directly.

The problem is to **analyze and transform this raw data into meaningful insights** that help identify:

- Trends over time
- Disaster-prone regions
- Most common disaster types
- Assistance distribution

This project aims to solve this by **developing visualizations and dashboards** that make disaster data easy to understand and useful for decision-making.

3. Project Objectives

- Identify temporal trends in disaster occurrences
- Analyze geographical distribution of disasters
- Study incident type patterns
- Understand relationship between disaster type and assistance
- Create clear and interactive visualizations
- Generate insights for better understanding

4. Tech Stack Used

- Python
- Pandas
- NumPy

- Matplotlib
- Seaborn
- Plotly
- Jupyter Notebook
- Power BI

5. Methodology / Workflow

1. Data Collection (FEMA dataset)
2. Data Cleaning (handling missing values, formatting)
3. Exploratory Data Analysis (EDA)
4. Visualization Creation
5. Insight Generation
6. Dashboard Development
7. Final Presentation

6. Milestone-wise Work Explanation

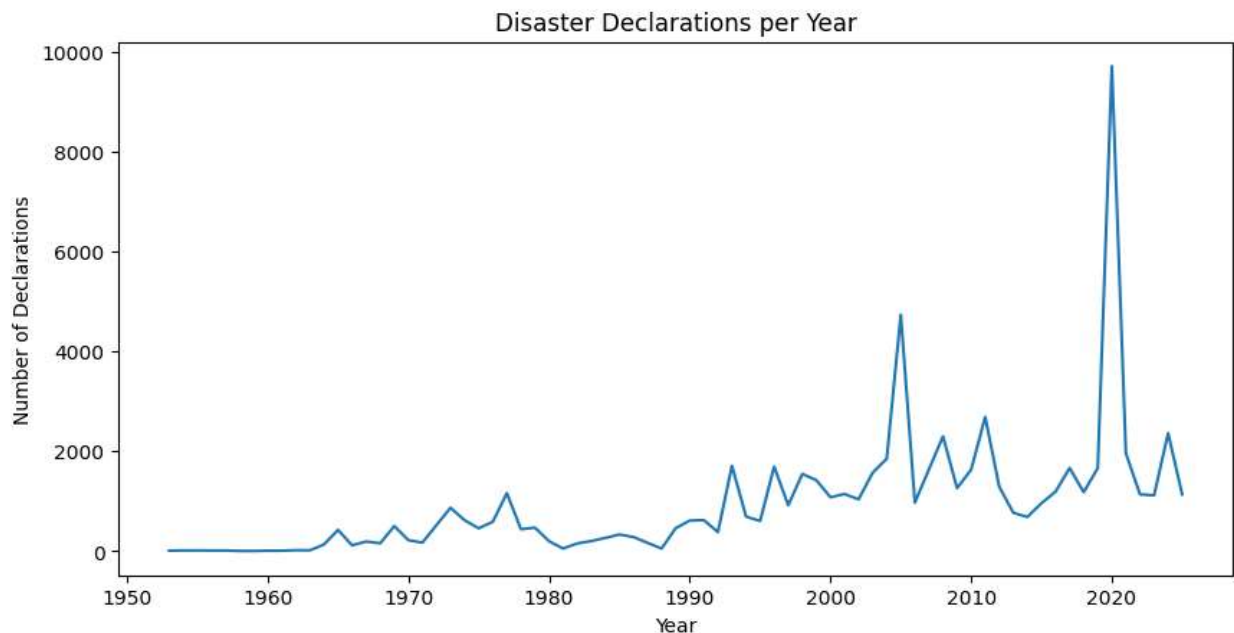
Milestone 1: Data Cleaning & Initial Analysis

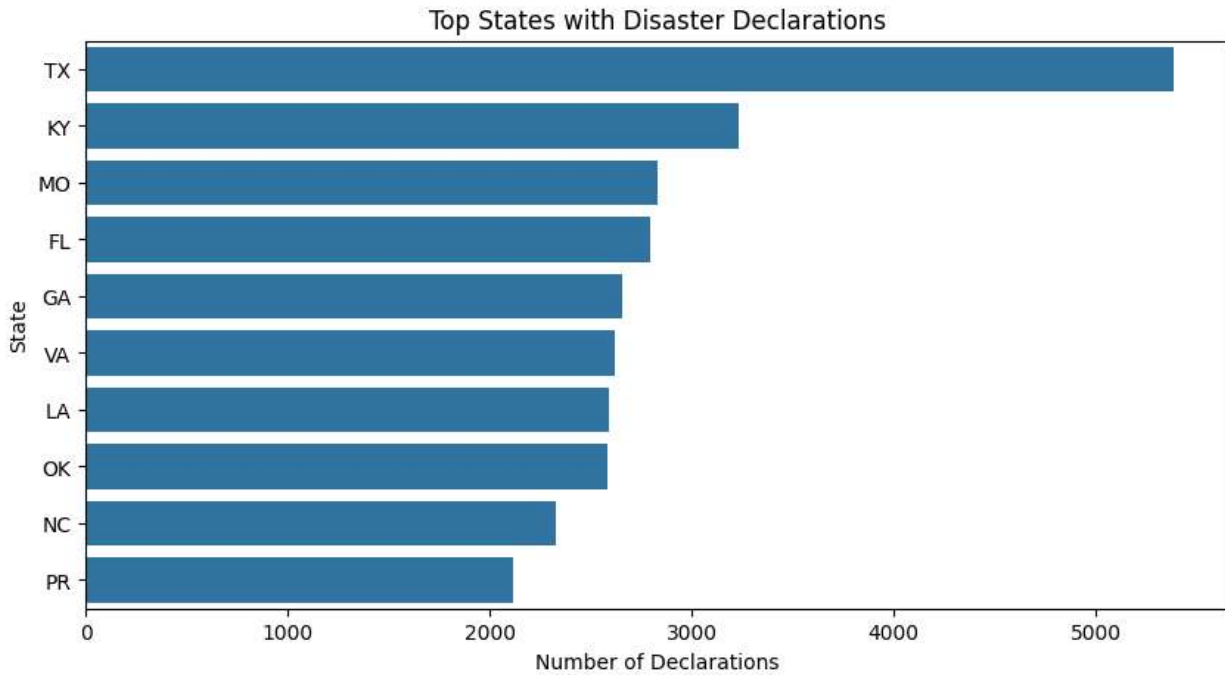
Work Done:

- Loaded FEMA dataset using Pandas
- Handled missing values
- Converted date formats
- Standardized categorical columns (incident type)
- Performed basic EDA

Outputs:

- Summary statistics
- Frequency counts (year, state)





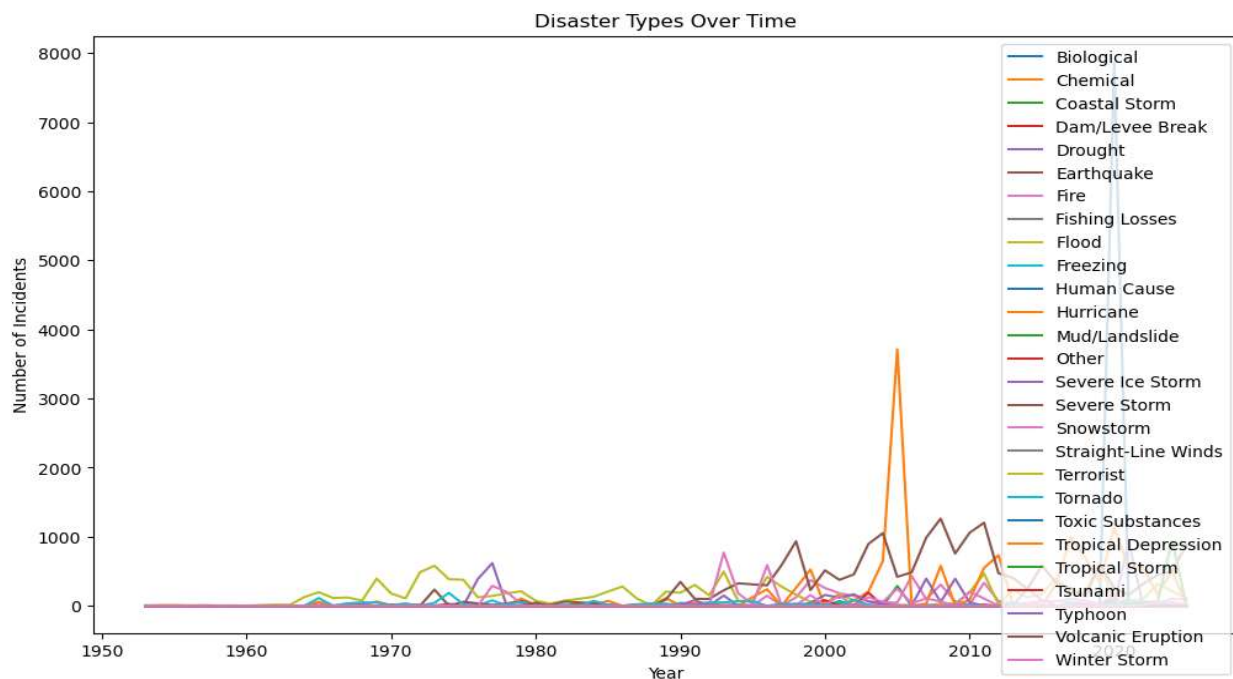
Milestone 2: Temporal Analysis

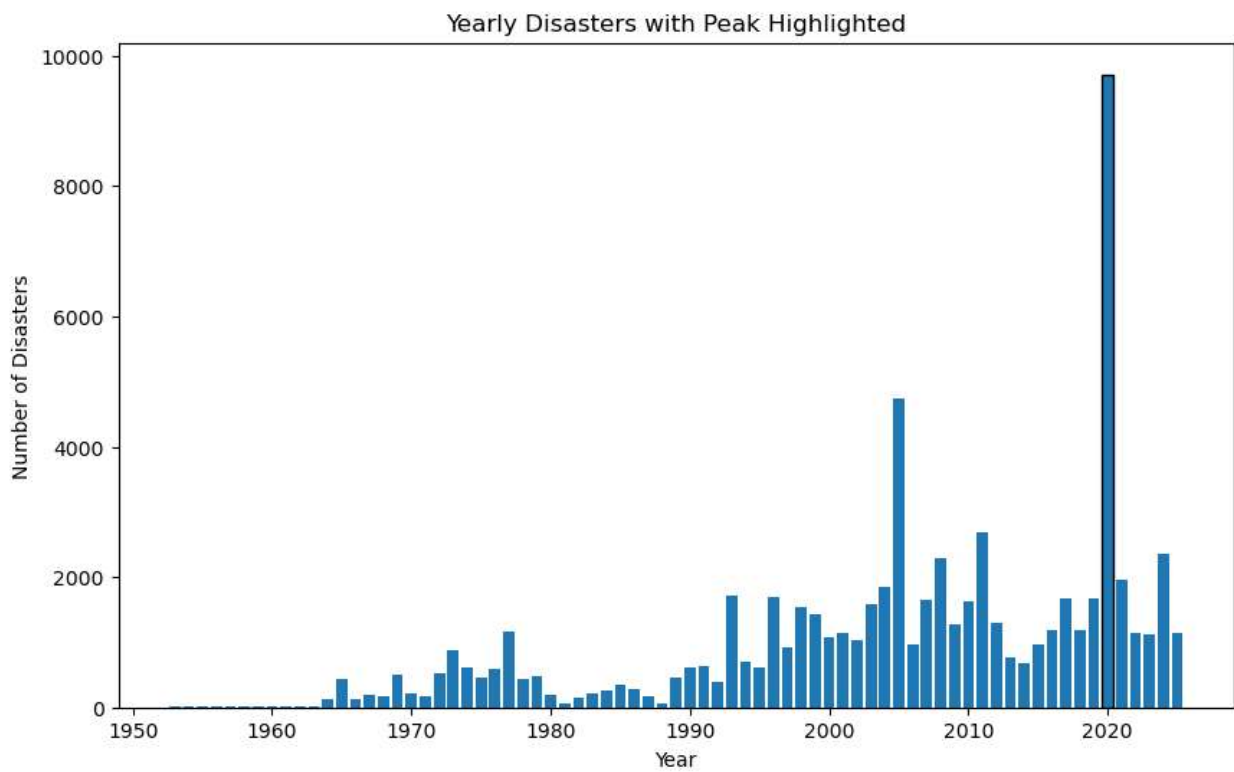
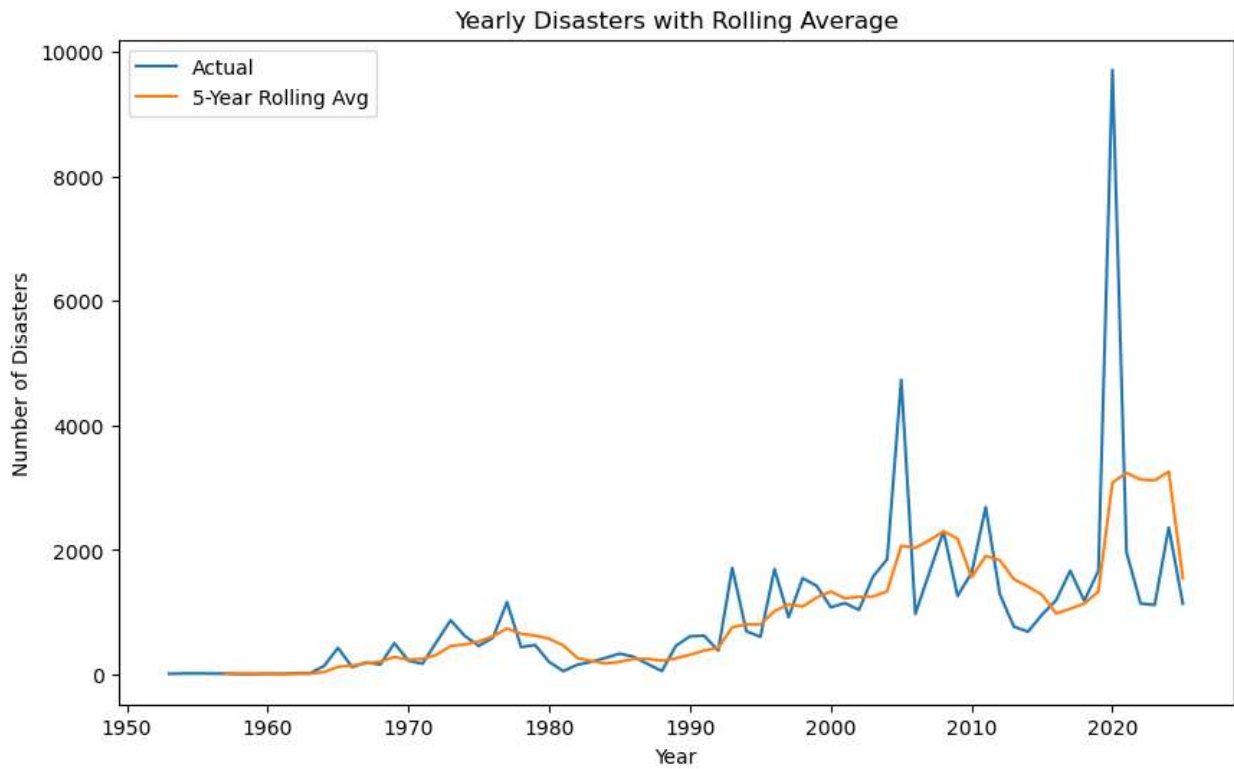
Work Done:

- Analyzed disaster trends over time
- Created time-series visualizations
- Studied patterns and spikes

Insights:

- Increase in disasters over years
- Sudden spikes indicate major events





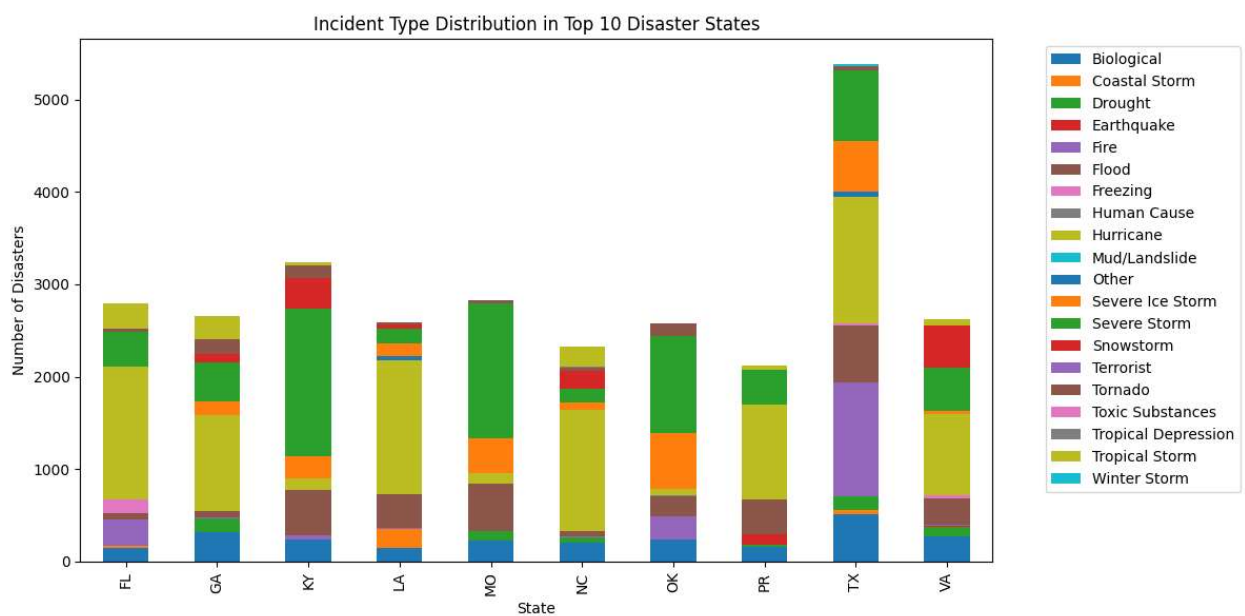
Milestone 3: Geographical Analysis

Work Done:

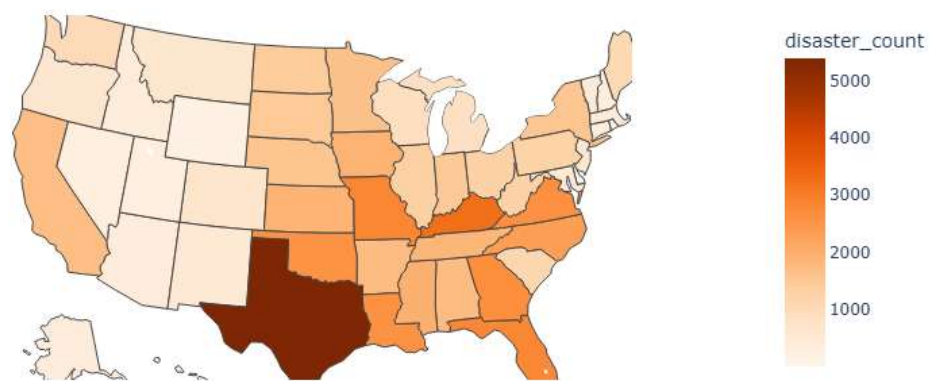
- Aggregated data by state
- Created map visualizations
- Compared disaster distribution

Insights:

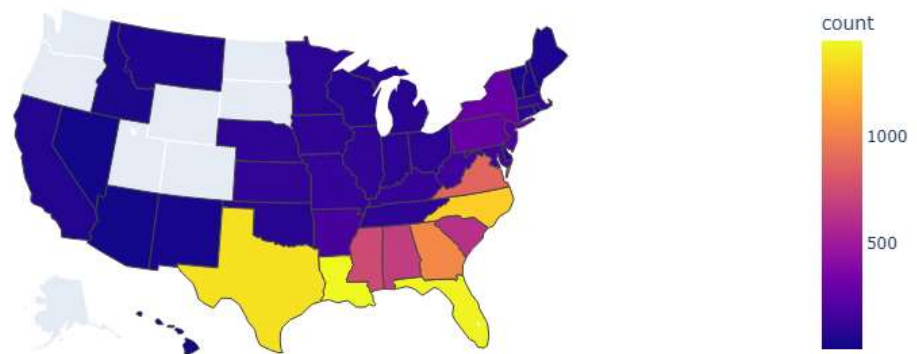
- Certain states are highly disaster-prone
- Regional patterns observed



US Disaster Declarations by State



Hurricane Disaster Hotspots



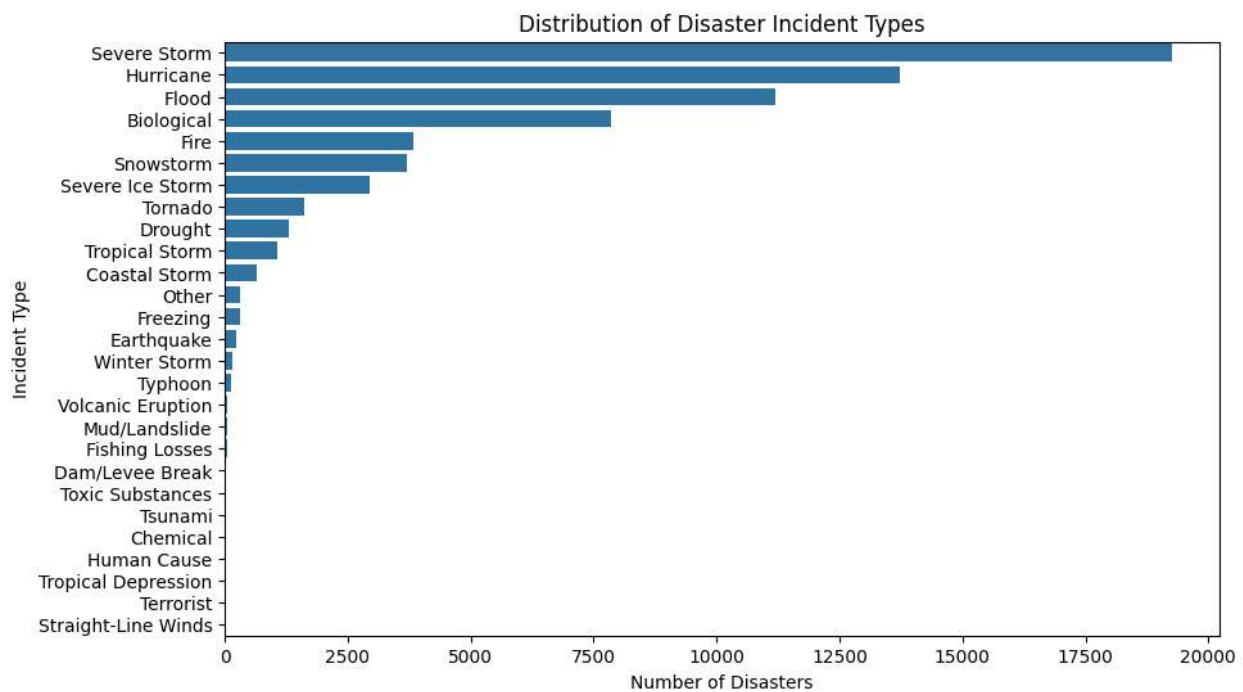
Milestone 4: Incident Type & Final Dashboard

Work Done:

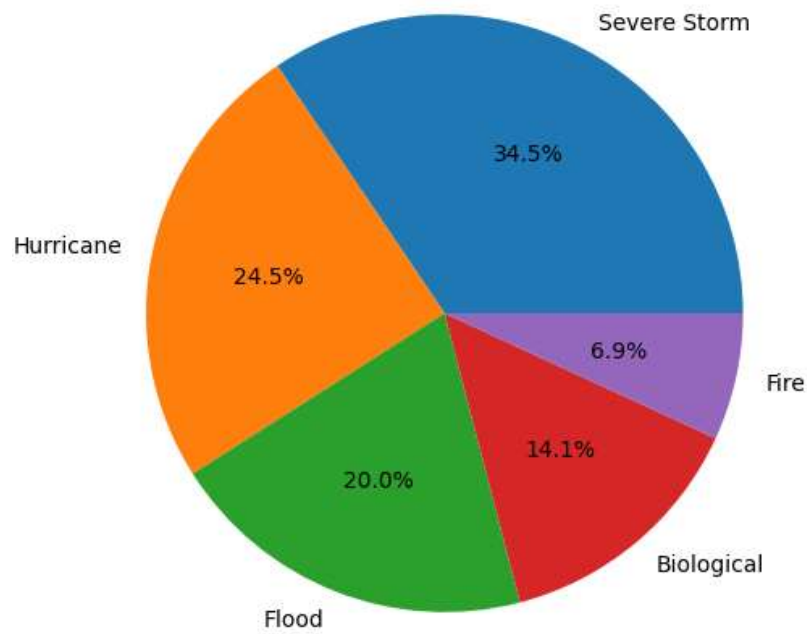
- Analyzed disaster types
- Compared assistance (IH vs PA)
- Built Power BI Dashboard

Insights:

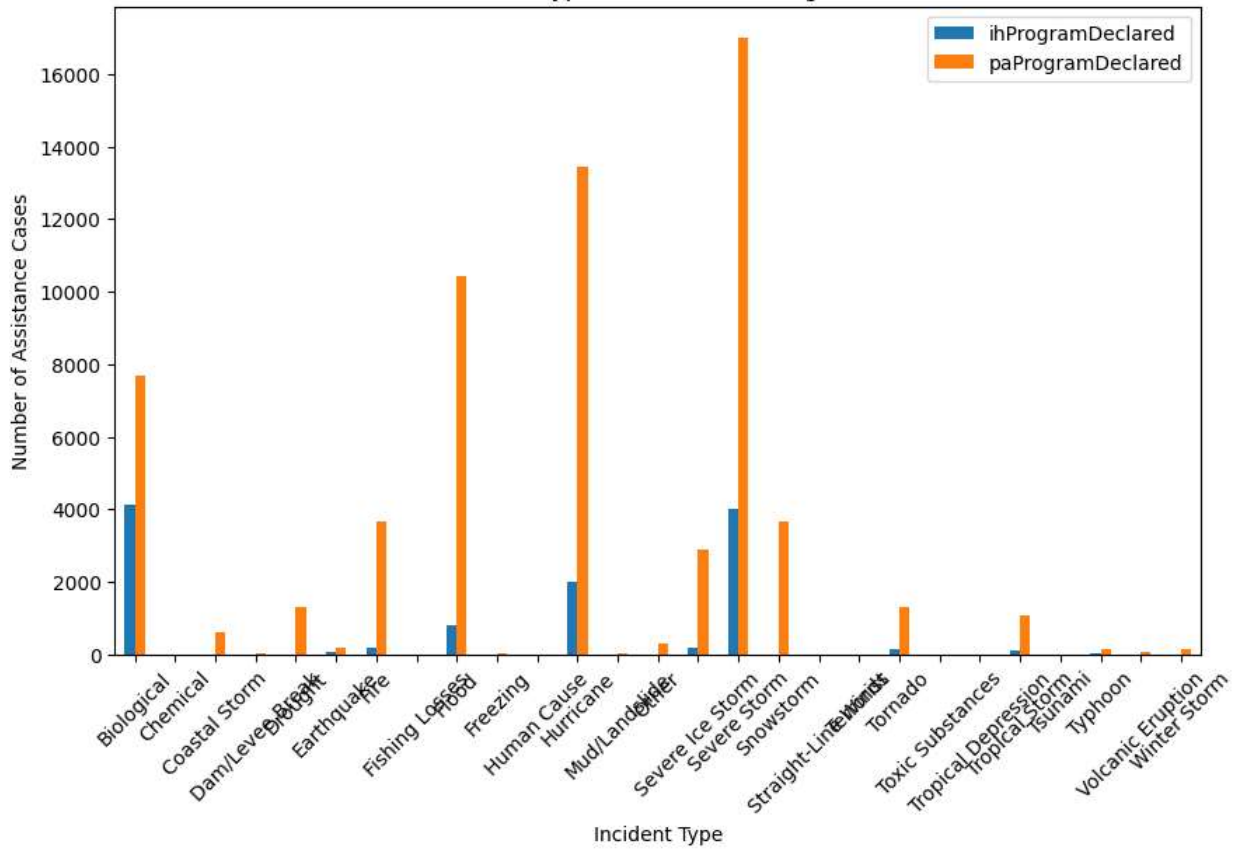
- Severe storms and floods are most common
- Assistance varies by disaster type

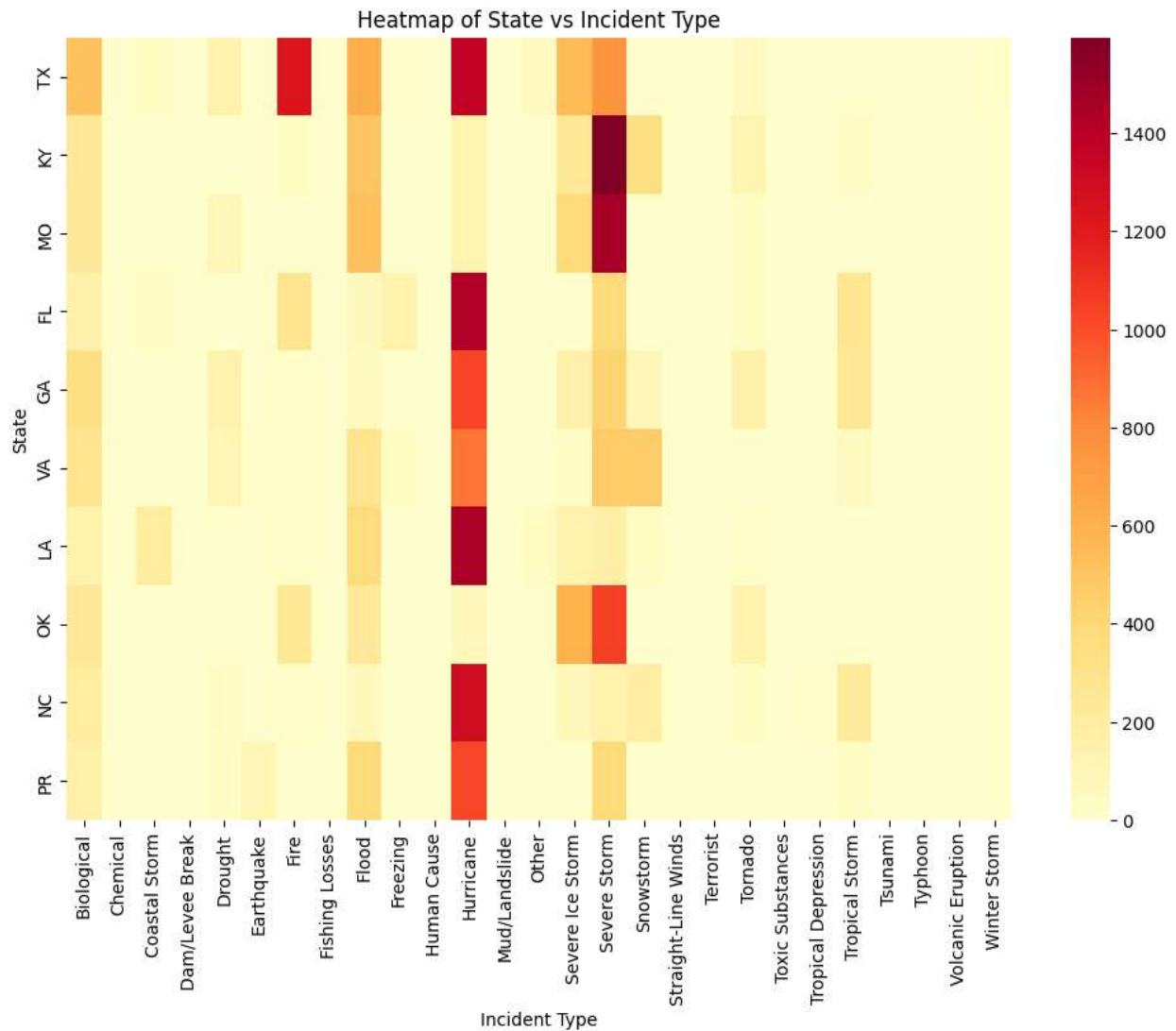


Top Disaster Types



Incident Type vs Assistance Programs



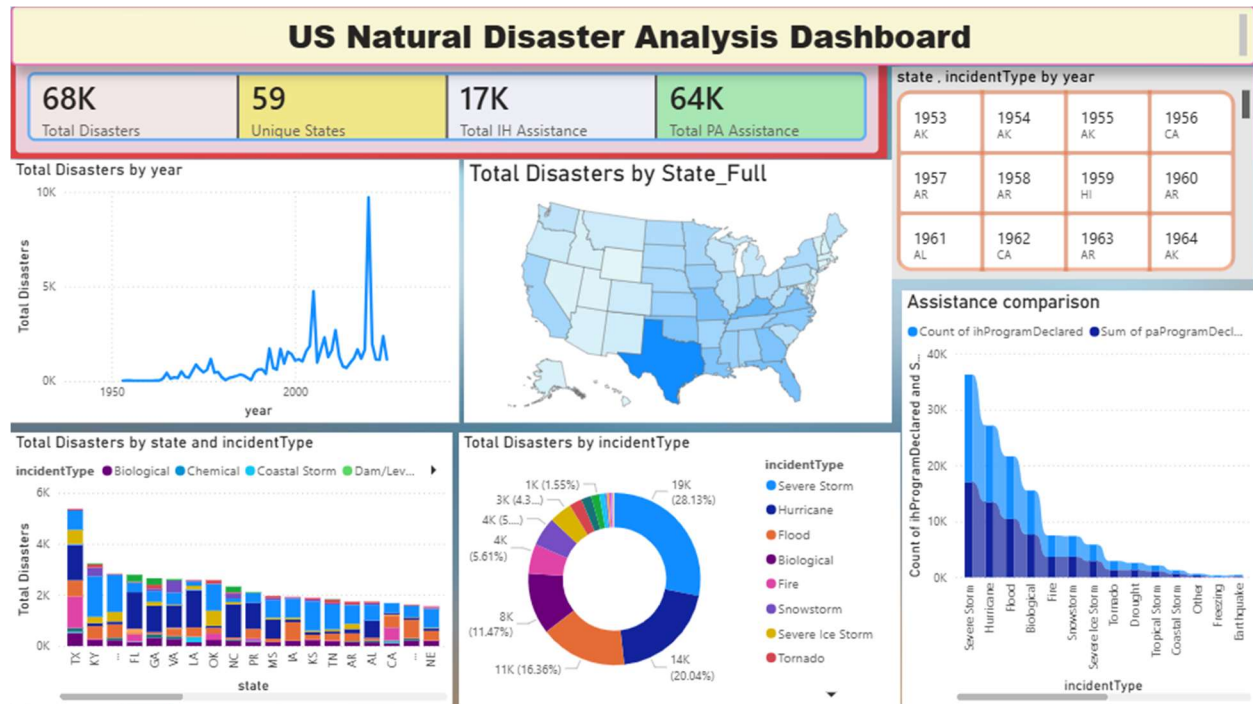


7. Dashboard Explanation

The dashboard provides:

- KPI overview (Total disasters, states, assistance)
- Year-wise trend analysis
- State-wise disaster distribution
- Incident type breakdown
- Assistance comparison

It allows users to **interactively explore disaster data** and identify key patterns.



8. Key Insights

- Disaster frequency is increasing over time
- Some states experience more disasters than others
- Severe storms are the most common disaster type
- Assistance allocation differs by disaster category

9. Project Outcomes & Learnings

- Learned data cleaning and preprocessing
- Gained experience in EDA
- Developed visualization skills
- Built an interactive Power BI dashboard
- Understood real-world data analysis workflow

10. Conclusion

This project successfully transformed raw FEMA disaster data into meaningful insights using data analysis and visualization techniques.

The dashboard helps in:

- Understanding disaster trends
- Identifying high-risk regions

- Supporting better decision-making

11. Future Scope

- Add real-time disaster data
- Predict future disasters using ML
- Improve dashboard interactivity
- Integrate weather datasets
- Deploy as a web application

<C:\Users\DELL\OneDrive\Desktop\data-science folder>

12.. GitHub Repository

The complete project files including datasets, Jupyter notebooks (Milestone 1–4), and the Power BI dashboard are available in the GitHub repository.

Repository Name: USND19

GitHub Link: https://github.com/GKSJ19/USND/tree/Shreyanka_YH?tab=readme-ov-file

This repository contains:

- Data cleaning and preprocessing (Milestone 1)
- Temporal analysis (Milestone 2)
- Geographical analysis (Milestone 3)
- Incident type analysis and final dashboard (Milestone 4)
- Power BI dashboard file (.pbix)